

DD_FUL-10 - Technology Migration Fulfillment

Version: v1.0 **Bundle:** 06_fulfillment **Status:** Draft for review **Primary owner:** Fulfillment / provisioning team **Related DDs:** DD_SUB-WF-MIGRATION-01, DD_WO-01, DD_WO-01-CONTRACT-FOR-SUB-WF, DD_FUL-03, DD_FUL-09, DD_PROV-INT-01, DD_RLM-CFG-01, DD_OSR-01, DD_OSR-INSTANCE-01, DD_OSR-02, DD_BIL-01, DD_EM-CFG-04, FOUNDATION_DROOLS

MVP simplification baseline: FUL-10 is phased after activation, payment, WO installation, OSR basics, and provisioning integration are stable. First implementation is admin-only cohort migration, not customer self-service cross-technology migration. FUL-10 calls WO only for field work and does not create a separate WO migration engine unless an operator process requires it.

1. Purpose

FUL-10 defines the fulfillment-side execution for a customer or network migration that changes the access technology used by an active subscription.

Examples:

- HFC to GPON migration;
- GPON to HFC migration where an operator supports both access networks;
- legacy access to fiber migration;
- operator-driven area cutover from one network topology to another;
- package migration that requires a different access technology or equipment profile.

SUB-WF-MIGRATION-01 owns the subscription operation: commercial validation, target package, target HomePass, billing policy, customer-facing operation state, and final subscription state update. FUL-10 owns the technical fulfillment: migration validation against provisioning/network constraints, work-order creation, equipment plan, source technology teardown, target technology activation, network reconciliation, and fulfillment outcome callback.

Use FUL-10 when:

- source and target HomePass technologies are different;
- the migration requires new access equipment or new network binding;
- the subscription migration flow needs a technical cutover before it can commit;
- an operator performs a planned technology retirement or area cutover;
- manual force-complete, rollback, or exception repair may be needed after network provisioning.

Procurement is related only upstream. OSR-02 replenishes stock before customer jobs start. FUL-10 must not call procurement during a migration. Runtime customer migrations consume equipment already available, reserved, or assigned through OSR/WO flows.

2. Scope And Boundaries

Capability	Owner
Subscription migration operation, package/homepass choice, customer-facing status	SUB-WF-MIGRATION-01
Technical migration fulfillment and cutover orchestration	FUL-10
Field visit, technician assignment, installation evidence	WO-01
Equipment replacement fulfillment	FUL-09
Network/service provisioning commands	FUL-03 / PROV-INT-01
HomePass topology and service availability	RLM-CFG-01
Stock balances and reservations	OSR-01
Serialized equipment state and binding	OSR-INSTANCE-01
Upstream replenishment and purchase orders	OSR-02

Capability	Owner
Charges, deposits, proration, migration fees	BIL-01 / BIL-02
Configurable risky override approval	EM-CFG-04
Operator-variable eligibility, cutover, and exception rules	Drools

Boundary rules:

- FUL-10 does not approve the commercial migration. SUB-WF-MIGRATION-01 must start or authorize the operation.
- FUL-10 does not directly update subscription master data. It calls back the migration outcome so SUB-WF can commit.
- FUL-10 does not schedule technicians directly if the WO module is enabled. It creates or references a WO.
- FUL-10 does not directly edit OSR stock or equipment instance tables. It uses OSR APIs and events.
- FUL-10 may call FUL-09 when the migration includes an equipment replacement/swap.
- FUL-10 may call FUL-03/PROV-INT to activate/deactivate access services.
- EM-CFG-04 is used only for approval policies such as force-complete, forced rollback, risky serial correction, or non-standard cutover. Camunda is involved only when the configured approval policy mode is BPMN.

3. Process Overview

```

flowchart LR
    A["SUB-WF migration requests technical fulfillment"] --> B["FUL-10 validates source/target technology"]
    B --> C["Build equipment and work plan"]
    C --> D["Create migration WO if field work required"]
    D --> E["Technician installs target access"]
    E --> F["FUL-10 executes source teardown and target activation"]
    F --> G["FUL-10 reconciles BSS and network state"]
    G --> H["FUL-10 sends outcome to SUB-WF"]
    H --> I["SUB-WF commits subscription migration"]

```

For an HFC to GPON migration:

1. SUB-WF-MIGRATION validates the target package and target HomePass.
2. FUL-10 validates that the target HomePass can serve the package on GPON.
3. FUL-10 creates a WO with migration context.
4. WO installs the ONT and captures serial, port, GPS, and evidence.
5. FUL-10 calls provisioning to activate GPON service.
6. FUL-10 tears down or suspends HFC service according to the cutover policy.
7. FUL-10 reconciles network state.
8. SUB-WF commits the subscription only after FUL-10 returns COMPLETED.

4. Status Model

Status	Meaning
REQUESTED	Technical migration fulfillment requested.
VALIDATED	Source and target technology context validated.
WO_CREATED	Work order created or linked.
EQUIPMENT_PLANNED	Required equipment and reuse/recovery plan decided.
FIELD_READY	Field installation evidence received and target path is ready for cutover.
CUTOVER_IN_PROGRESS	Source teardown and target activation are running.
RECONCILING	Provisioning/network state is being verified.
COMPLETED	Technical migration completed and callback sent.
FAILED	Technical migration failed and needs retry or repair.
PENDING_APPROVAL	Risky manual action requires EM-CFG-04 approval.
ROLLBACK_IN_PROGRESS	Approved rollback is being executed.

Status	Meaning
CANCELLED	Fulfillment cancelled before irreversible technical changes.

5. Data Model

5.1 technology_migration_fulfillment

What it is for: stores one technical fulfillment execution for a cross-technology migration.

When to use it: create one row when SUB-WF-MIGRATION asks fulfillment to execute a migration where source and target access technology differ, or where network/equipment migration must be orchestrated separately from normal same-tech shifting.

Sample meaning: tmf-001 is the technical fulfillment that migrates subscription SUB-700045 from HFC HomePass HP-HFC-0100 to GPON HomePass HP-GPON-0440.

Column	Type	Purpose	Sample
migration_fulfillment_id	TEXT PK	Technical fulfillment id.	tmf-001
operator_code	TEXT	Tenant/operator.	WIK
subscription_operation_id	TEXT	SUB-WF migration operation id.	op-mig-001
subscription_id	TEXT	Subscription being migrated.	SUB-700045
account_id	TEXT	Billing/account id.	ACC-900045
customer_id	TEXT	Customer id.	CUS-100045
source_technology	TEXT	Current access technology.	HFC
target_technology	TEXT	Target access technology.	GPON
source_homepass_id	TEXT	Current HomePass.	HP-HFC-0100
target_homepass_id	TEXT	Target HomePass.	HP-GPON-0440
migration_type	TEXT	ACCESS_TECH_CHANGE, NETWORK_UPGRADE, AREA_CUTOVER, PACKAGE_TECH_REQUIREMENT.	ACCESS_TECH_CHANGE
status	TEXT	Technical fulfillment state.	CUTOVER_IN_PROGRESS
wo_id	TEXT NULL	WO used for field execution.	wo-mig-001
equipment_swap_fulfillment_id	TEXT NULL	FUL-09 execution if equipment swap/rebind is required.	esf-010
approval_request_id	TEXT NULL	EM-CFG-04 request for risky action.	apr-tmf-001
created_at	TIMESTAMP	Creation timestamp.	2026-08-03T08:00:00Z
completed_at	TIMESTAMP NULL	Completion timestamp.	null

Recommended indexes:

- unique (operator_code, subscription_operation_id).
- (operator_code, subscription_id, status).
- (wo_id).
- (approval_request_id).

5.2 technology_migration_validation_check

What it is for: stores validation checks performed before technical migration execution.

When to use it: record each check that decides if the migration can proceed, needs approval, or must be rejected before field/provisioning work starts.

Sample meaning: tmvc-001 confirms that target HomePass HP-GPON-0440 supports internet and voice for the target package.

Column	Type	Purpose	Sample
validation_check_id	TEXT PK	Validation row id.	tmvc-001
migration_fulfillment_id	TEXT	Parent fulfillment.	tmf-001

Column	Type	Purpose	Sample
check_code	TEXT	Validation name.	TARGET_TECH_SUPPORTS_PACKAGE
check_group	TEXT	HOMEPASS, PACKAGE, CAPACITY, EQUIPMENT, BILLING, PROVISIONING.	HOMEPASS
expected_value_json	JSONB	Expected condition.	{"requiredFamilies":["INTERNET", "VOICE"]}
actual_value_json	JSONB NULL	Actual condition found.	{"supportedFamilies":["INTERNET", "VOICE", "TV"]}
status	TEXT	PASS, WARN, FAIL, APPROVAL_REQUIRED.	PASS
message	TEXT	Human-readable result.	Target GPON HomePass supports required services.
checked_at	TIMESTAMP	Check timestamp.	2026-08-03T08:01:00Z

Recommended indexes:

- (migration_fulfillment_id, status).
- (check_code, status).

5.3 technology_migration_equipment_plan

What it is for: stores the equipment decision for the migration.

When to use it: create rows for equipment that must be kept, recovered, newly installed, swapped, or re-bound as part of the migration.

Sample meaning: tmep-001 says the old HFC modem must be recovered and a new GPON ONT must be installed.

Column	Type	Purpose	Sample
equipment_plan_id	TEXT PK	Equipment plan row id.	tmep-001
migration_fulfillment_id	TEXT	Parent fulfillment.	tmf-001
plan_role	TEXT	SOURCE_RECOVERY, TARGET_INSTALL, REUSE, RETURN_TO_STOCK, CUSTOMER_KEEP.	TARGET_INSTALL
sku_id	TEXT NULL	Required SKU.	WIK-ONT-HUAWEI-EG8145V5
source_instance_id	TEXT NULL	Existing equipment instance.	inst-hfc-001
target_instance_id	TEXT NULL	New/target equipment instance.	inst-gpon-001
reservation_ref	TEXT NULL	OSR reservation or WO stock reference.	res-ont-001
fulfillment_action	TEXT	NO_ACTION, CALL_FUL_09, WO_INSTALL_ONLY, RECOVER_ONLY.	CALL_FUL_09
status	TEXT	PLANNED, RESERVED, ASSIGNED, COMPLETED, FAILED.	RESERVED
created_at	TIMESTAMP	Creation timestamp.	2026-08-03T08:02:00Z

Recommended indexes:

- (migration_fulfillment_id, status).
- (target_instance_id).
- (reservation_ref).

5.4 technology_migration_cutover_step

What it is for: stores each technical command/check executed during source teardown, target activation, reconciliation, or rollback.

When to use it: create one row per provisioning command or technical checkpoint sent through PROV-INT/FUL-03 or another approved adapter.

Sample meaning: tmcs-001 records the command that activates the GPON ONT for subscription SUB-700045.

Column	Type	Purpose	Sample
cutover_step_id	TEXT PK	Step id.	tmcs-001
migration_fulfillment_id	TEXT	Parent fulfillment.	tmf-001
step_order	INT	Execution order.	20
step_code	TEXT	Step name.	GPON_ACTIVATE_TARGET_ONT
target_system	TEXT	Provisioning/network target.	OLT_NMS
command_ref	TEXT NULL	PROV-INT command id.	prov-cmd-9001
status	TEXT	PENDING, SENT, SUCCESS, FAILED, SKIPPED, COMPENSATED.	SUCCESS
request_json	JSONB	Command request snapshot.	{"serial": "ONT123456"}
response_json	JSONB NULL	Command response snapshot.	{"result": "OK"}
failure_code	TEXT NULL	Failure code if failed.	null
started_at	TIMESTAMP NULL	Start time.	2026-08-03T10:30:00Z
completed_at	TIMESTAMP NULL	Completion time.	2026-08-03T10:30:15Z

Recommended indexes:

- (migration_fulfillment_id, step_order).
- (migration_fulfillment_id, status).
- (command_ref).

5.5 technology_migration_exception

What it is for: stores technical exceptions and manual recovery cases for a migration fulfillment.

When to use it: create one row when validation, provisioning, reconciliation, rollback, or force-complete needs investigation or approval.

Sample meaning: tmex-001 records that GPON activation succeeded but HFC source teardown failed, so NOC must decide whether to retry teardown or force-complete.

Column	Type	Purpose	Sample
exception_id	TEXT PK	Exception id.	tmex-001
migration_fulfillment_id	TEXT	Parent fulfillment.	tmf-001
exception_type	TEXT	VALIDATION_FAILED, PROVISIONING_FAILED, NETWORK_MISMATCH, ROLLBACK_REQUIRED, FORCE_COMPLETE.	PROVISIONING_FAILED
severity	TEXT	LOW, MEDIUM, HIGH, CRITICAL.	HIGH
status	TEXT	OPEN, PENDING_APPROVAL, APPROVED, RESOLVED, CANCELLED.	OPEN
approval_request_id	TEXT NULL	EM-CFG-04 approval request.	apr-tmf-001
assigned_to_team_code	TEXT NULL	Team responsible for repair.	NOC_GPON
created_at	TIMESTAMP	Creation timestamp.	2026-08-03T10:35:00Z
resolved_at	TIMESTAMP NULL	Resolution timestamp.	null

Recommended indexes:

- (migration_fulfillment_id, status).
- (approval_request_id).
- (assigned_to_team_code, status).

6. APIs

6.1 Start Technical Migration Fulfillment

Called by SUB-WF-MIGRATION or a trusted backoffice orchestration API after the migration operation has been accepted.

```
POST /api/technology-migrations/fulfillments
Idempotency-Key: tmf-op-mig-001
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "subscriptionOperationId": "op-mig-001",
  "subscriptionId": "SUB-700045",
  "accountId": "ACC-900045",
  "customerId": "CUS-100045",
  "sourceTechnology": "HFC",
  "targetTechnology": "GPON",
  "sourceHomepassId": "HP-HFC-0100",
  "targetHomepassId": "HP-GPON-0440",
  "targetPackageRef": "WIK-FIBER-50M",
  "migrationType": "ACCESS_TECH_CHANGE"
}
```

Response:

```
{
  "migrationFulfillmentId": "tmf-001",
  "status": "REQUESTED"
}
```

Rules:

- Same operatorCode + subscriptionOperationId must return the existing fulfillment.
- If the same subscription has another active technology migration fulfillment, return 409.
- Validate that the caller is an approved internal module or backoffice role.

6.2 Validate Migration

```
POST /api/technology-migrations/fulfillments/tmf-001/validate
```

The API runs Drools eligibility rules and stores validation check rows.

6.3 Create Or Link Work Order

```
POST /api/technology-migrations/fulfillments/tmf-001/create-work-order
Content-Type: application/json
```

```
{
  "preferredScheduleFrom": "2026-08-04T08:00:00Z",
  "preferredScheduleTo": "2026-08-04T13:00:00Z",
  "requiresCustomerVisit": true
}
```

FUL-10 calls WO-01:

```
POST /api/work-orders
Content-Type: application/json
```

```
{
  "operatorCode": "WIK",
  "kind": "TECHNOLOGY_MIGRATION",
  "originatingContextType": "TECHNOLOGY_MIGRATION_FULFILLMENT",
  "originatingContextId": "tmf-001",
  "subscriptionId": "SUB-700045",
  "accountId": "ACC-900045",
  "homepassId": "HP-GPON-0440",
  "metadata": {
    "sourceTechnology": "HFC",
    "targetTechnology": "GPON",
    "targetPackageRef": "WIK-FIBER-50M"
  }
}
```

6.4 Execute Cutover

```
POST /api/technology-migrations/fulfillments/tmf-001/execute-cutover
Content-Type: application/json
```

```
{
  "fieldCompletionRef": "wo-mig-001",
  "targetEquipmentInstanceId": "inst-gpon-001",
  "cutoverMode": "MAKE_BEFORE_BREAK"
}
```

6.5 Outcome Callback To SUB-WF

POST /internal/subscription-migrations/op-mig-001/fulfillment-outcome
Content-Type: application/json

```
{
  "migrationFulfillmentId": "tmf-001",
  "status": "COMPLETED",
  "sourceTechnology": "HFC",
  "targetTechnology": "GPON",
  "completedAt": "2026-08-04T11:30:00Z"
}
```

6.6 Request Approval For Risky Action

FUL-10 calls EM-CFG-04 when configured policy requires approval.

POST /api/approval-requests
Content-Type: application/json

```
{
  "operatorCode": "WIK",
  "approvalCode": "TECH_MIGRATION_FORCE_COMPLETE",
  "originRefType": "TECHNOLOGY_MIGRATION_FULFILLMENT",
  "originRefId": "tmf-001",
  "requestedByUserId": "usr-noc-lead",
  "payload": {
    "exceptionId": "tmex-001",
    "reason": "Target GPON verified manually; source HFC teardown retry will run later."
  }
}
```

7. Rules And Drools

Use Drools for rules that vary by operator, region, technology, or commercial policy.

Recommended packages:

- rules.fulfillment.technology_migration.eligibility
- rules.fulfillment.technology_migration.equipment
- rules.fulfillment.technology_migration.cutover
- rules.fulfillment.technology_migration.exception

Example rules:

```
package rules.fulfillment.technology_migration.eligibility

rule "Reject migration when target technology does not support package services"
when
  $f : TechnologyMigrationFact(targetSupportsAllRequiredServices == false)
  $d : TechnologyMigrationDecision()
then
  $d.reject("TARGET_TECH_PACKAGE_INCOMPATIBLE");
end

package rules.fulfillment.technology_migration.cutover

rule "HFC to GPON make-before-break"
when
  $f : TechnologyMigrationFact(sourceTechnology == "HFC", targetTechnology == "GPON")
  $d : TechnologyMigrationDecision()
then
  $d.setCutoverMode("MAKE_BEFORE_BREAK");
  $d.addStep("GPON_ACTIVATE_TARGET_ONT");
  $d.addStep("GPON_RECONCILE_TARGET");
  $d.addStep("HFC_SUSPEND_SOURCE_MODEM");
  $d.addStep("HFC_RECONCILE_SOURCE");
end

package rules.fulfillment.technology_migration.exception

rule "Force complete requires approval when source teardown failed"
when
  $f : TechnologyMigrationExceptionFact(exceptionType == "PROVISIONING_FAILED",
                                         targetActivationVerified == true,
                                         sourceTeardownVerified == false)
  $d : TechnologyMigrationExceptionDecision()
```

```

then
    $d.requireApproval("TECH_MIGRATION_FORCE_COMPLETE");
    $d.assignTeam("NOC_ACCESS");
    $d.setSeverity("HIGH");
end

```

Fixed validations such as required fields, idempotency, enum values, and foreign reference format belong in application code, not Drools.

8. Events

Publish events through the transactional outbox after the local database commit.

Event	When emitted	Key fields
TechnologyMigrationFulfillmentRequeste d	Fulfillment row created.	migrationFulfillmentId, subscriptionOperationId, subscriptionId
TechnologyMigrationValidated	Validation completes.	status, failedChecks
TechnologyMigrationWorkOrderCreated	WO is created/linked.	woId, migrationFulfillmentId
TechnologyMigrationCutoverStarted	Cutover begins.	cutoverMode, sourceTechnology, targetTechnology
TechnologyMigrationCompleted	Technical fulfillment completes.	completedAt, targetHomepassId
TechnologyMigrationFailed	Technical fulfillment fails.	failureCode, exceptionId
TechnologyMigrationExceptionOpened	Exception row created.	exceptionType, severity, assignedTeam

9. Frontend And Module Consumers

Consumer	What it uses
Backoffice migration console	Start/validate/status APIs, validation checks, exception state
Contractor technical mobile app	WO-01 migration task, equipment capture, field completion
NOC/support console	Cutover steps, reconciliation, exception repair
Subscription workflow	Start fulfillment and receive outcome callback
Reporting	Events and fulfillment tables for migration SLA/failure reporting

Frontend screens must show module boundaries clearly:

- commercial/package decision comes from SUB-WF and catalog APIs;
- field task status comes from WO APIs;
- equipment serials and stock come from OSR APIs;
- technical cutover and reconciliation come from FUL-10 APIs.

10. Infrastructure Recommendations

For a medium budget-conscious deployment:

- Run FUL-10 APIs in the existing fulfillment service deployment.
- Run FUL-10 logical workers in the shared fulfillment worker deployment; keep metrics by topic/handler.
- Use the fulfillment PostgreSQL schema for FUL-10 tables.
- Use the existing Kafka/outbox stack for events.
- Use existing PROV-INT adapters for network commands; do not introduce direct NMS calls in business code.
- Keep Camunda optional for this DD. Use it when the operator wants a visible long-running technical workflow. Use EM-CFG-04 for approvals regardless of whether the approval itself is simple or BPMN-backed.

Recommended logical worker handlers:

Handler	Purpose
ful10-validate-migration	Runs validation checks and Drools eligibility.
ful10-create-wo	Creates/links WO-01 migration work order.
ful10-plan-equipment	Builds equipment plan from OSR and package requirements.
ful10-execute-cutover	Sends provisioning commands through PROV-INT/FUL-03.
ful10-reconcile	Verifies network/BSS state.
ful10-handle-exception	Routes exception and calls EM-CFG-04 when needed.

These are logical handlers. One worker process may implement multiple handlers.

11. Test Scenarios

Minimum tests:

1. Start fulfillment is idempotent for the same subscription operation.
2. Second active technology migration for same subscription returns 409.
3. HFC to GPON validation passes when target HomePass supports package services.
4. Validation fails when target HomePass lacks required service family.
5. WO is created with kind=TECHNOLOGY_MIGRATION and correct origin context.
6. Equipment plan calls FUL-09 only when equipment swap/rebind is required.
7. Cutover executes target activation before source teardown for make-before-break policy.
8. Reconciliation failure creates `technology_migration_exception`.
9. Force-complete calls EM-CFG-04 when configured.
10. Completed fulfillment calls SUB-WF outcome endpoint once.

12. Implementation Notes

- Treat FUL-10 as the missing technical satellite for cross-technology migration.
- Keep the customer subscription active on the source technology until target activation has passed reconciliation, unless the operator explicitly configures break-before-make.
- Do not update subscription package/homepass from FUL-10. SUB-WF commits after receiving the success outcome.
- If a field visit is required, WO-01 remains the owner of assignment, technician evidence, and field finalization.
- If equipment changes are required, FUL-10 coordinates with FUL-09 instead of duplicating equipment swap provisioning logic.
- If procurement is missing stock, the operational answer is to stop before runtime fulfillment and let OSR replenishment happen through OSR-02. Do not insert purchase order logic into FUL-10.